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Three-Dimensional Integrative Model (Thermal, Technical, and Juridical) for Blockchain-Enabled Dual-Temperature Micro Cold-Storage Nodes in the TNBBS Corridor

Ilham Pradani [1, *]
[1] Research & Development Division, PT CHAIN ASIA CONECT, Jakarta & Krui, Pesisir Barat, Indonesia
Corresponding Author: r-d@chainasiaconnect.co.id | ORCID: 0009-0008-1310-6169

ABSTRACT
Background: Smallholder fishing communities in Krui face post-harvest losses up to 35% due to inadequate cold storage and informal dispute resolution mechanisms. Objectives: To synthesize a 3D framework encompassing thermodynamic enthalpy, edge hardware reliability, and Civil Code contractual certainty. Methods: Triangulated field instrumentation, thermodynamic cooling load calculations, and statutory contract modeling. Results: Demonstrates that technical compliance without contractual certainty leads to 60% higher operational default; integrating verifiable on-chain thermal proofs eliminates evidentiary ambiguity in regional commercial arbitration. Conclusion: Synthesizing law with thermodynamics creates durable physical-digital business infrastructure in frontier coastal communities.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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1. Introduction

1.1 Empirical Background

The Bromo-Batu-Sengguruh National Park (TNBBS) region covers more than **2,000 km²** and is inhabited by **3.4 million** people, the majority of whom reside in remote (3T) areas. Deficiencies in energy and transportation infrastructure lead to post-harvest losses of up to **27 %** (FAO, 2022). At the same time, stringent conservation regulations prohibit the construction of large-scale cooling facilities that could disturb the ecosystem.

1.2 Literature Review

Several previous studies have examined cold-chain logistics in 3T regions (Kumar & Singh, 2020; Lestari et al., 2021). However, these models are **centralised**, relying on a single temperature zone (± 4 °C) and emphasizing economic aspects while neglecting the thermal dynamics of micro-nodes or a collaborative legal framework. Blockchain-based temperature-monitoring research remains at a **conceptual stage** (Al-Shamsi et al., 2021) and has not yet been tested in a conservation context.

1.3 State of the Art & Research Gap

- **Centralisation vs. Decentralisation:** Traditional systems depend on a central warehouse and are inflexible to temperature variability during micro-node transit.
- **Single-Zone Temperature:** Most models assume a single temperature zone, whereas horticultural products and meat require **dual-temperature** conditions (e.g., 0–4 °C and −18 °C).
- **Legal-Technical Integration:** The literature remains fragmented across environmental regulations, renewable-energy law, and smart-legal contracts.
- **Empirical Validation:** No field-based experiment has simultaneously integrated thermal, technical, and juridical data.

1.4 Novelty of the Research

This study introduces an **Integrative Three-Dimensional Model** that unifies:

1. **Thermal Sub-Model:** Dynamic simulation of micro-node temperatures using CFD-IoT techniques.
2. **Technical Sub-Model:** Optimisation of a solar-mini-grid, Li-$FePO_4$ batteries, and blockchain communication protocols.
3. **Juridical Sub-Model:** Smart-contract legal code that encodes TNBBS regulations, HACCP standards, and energy-transaction provisions.

The model yields a **dynamic transit-scheduling algorithm** that minimises food loss, carbon footprint, and transaction costs.

1.5 Problem Statement

1. **Design Question:** How can a dual-temperature cold-storage micro-node be engineered to operate autonomously within a conservation zone?
2. **Integration Question:** How can TNBBS regulations be incorporated into a smart-contract framework?

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RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

Originality & Ethics Statement (COPE Standard): This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.24028633). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

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